

미션1

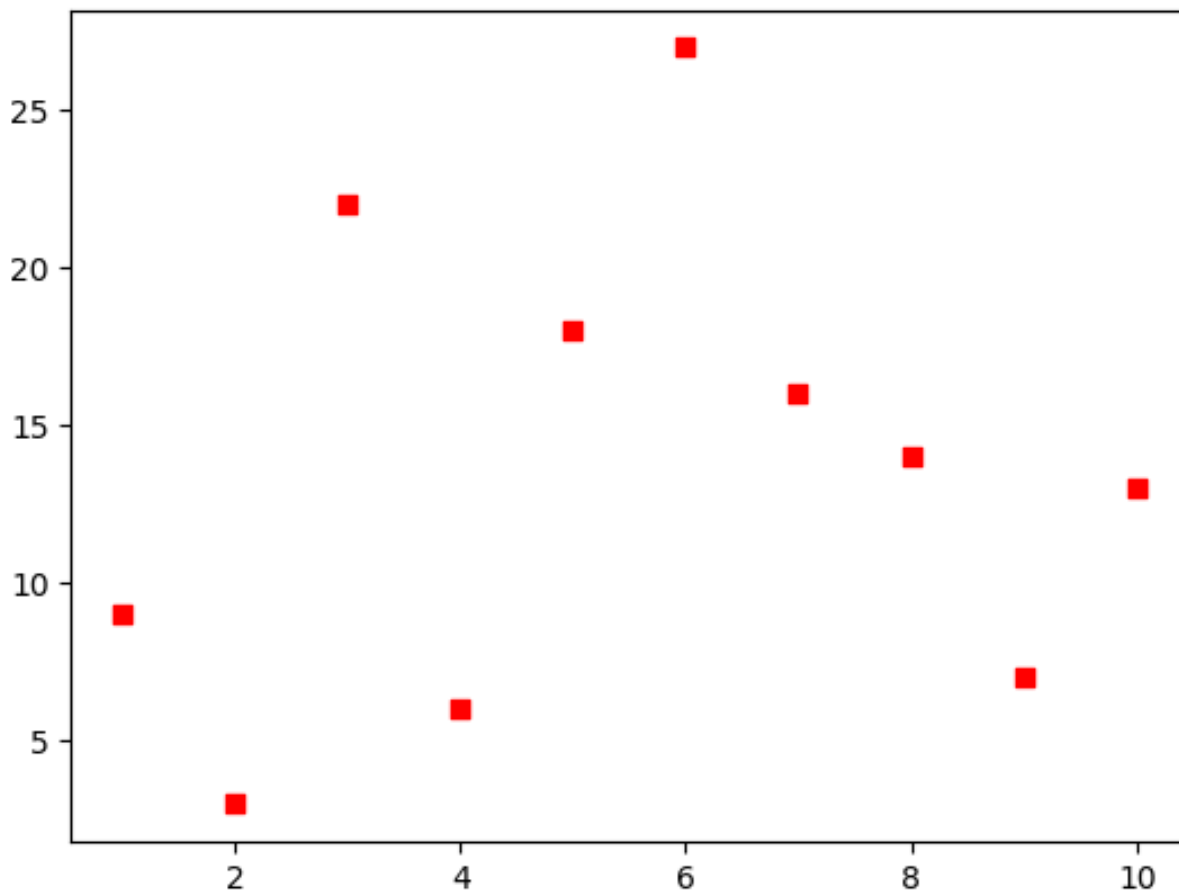
In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
```

```
x=np.arange(1,11)
y=np.random.randint(1,30,10)
```

```
print('x:',x)
print('y:',y)
plt.scatter(x,y,c='r',marker='s')
plt.show()
```

```
x: [ 1  2  3  4  5  6  7  8  9 10]
y: [ 9  3 22  6 18 27 16 14  7 13]
```



미션 2

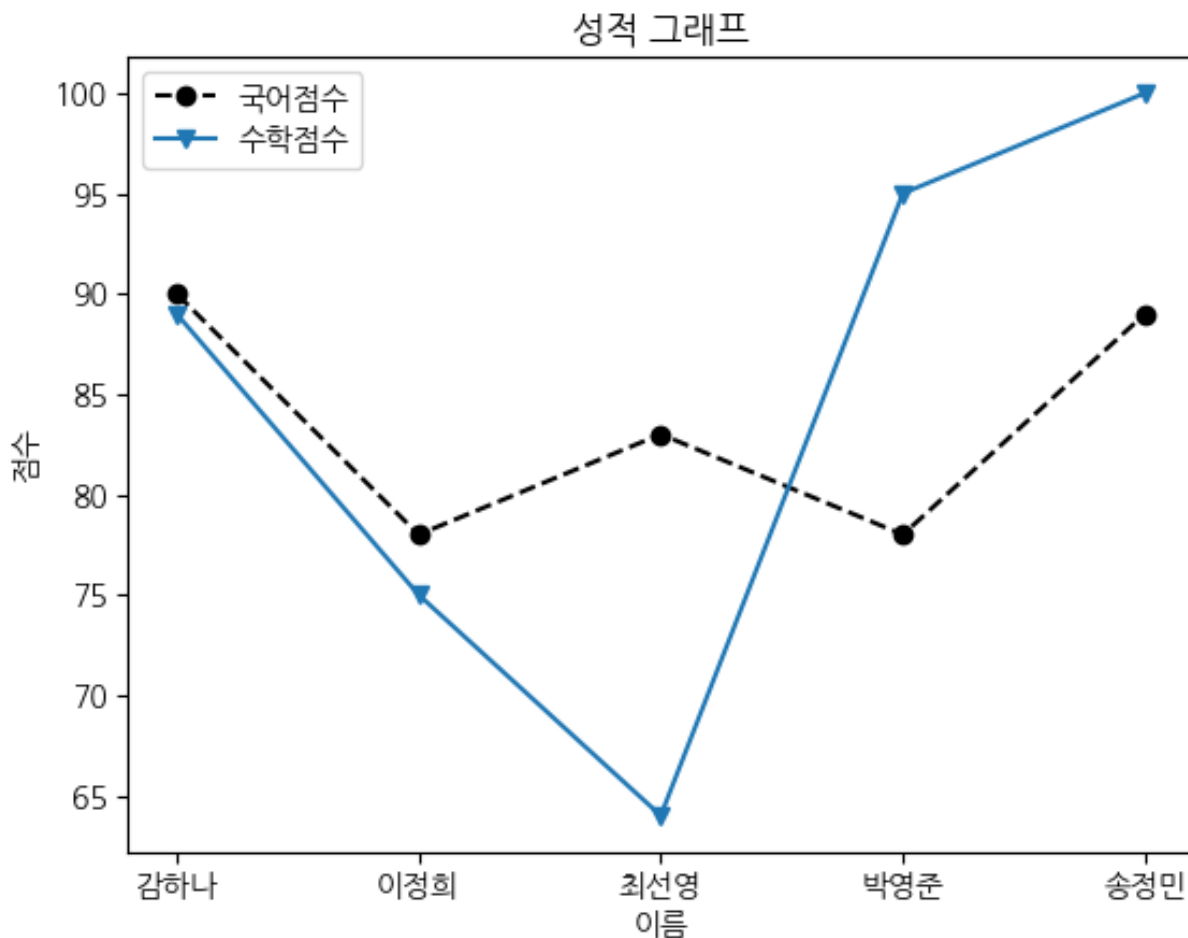
In [2]:

#코랩용 : 그래프의 선 모양, 색깔, 마커 모양은 아래 코드와 달라도 됨

```
import matplotlib.pyplot as plt
import pandas as pd
```

```
plt.rc('font', family='NanumGothic')
df=pd.read_excel('data1.xlsx')
```

```
plt.plot(df['name'], df['kor'], 'k--', marker='o', label='국어점수')
plt.plot(df['name'], df['math'], marker='v', label='수학점수')
plt.title('성적 그래프')
plt.xlabel('이름')
plt.ylabel('점수')
plt.legend()
plt.show()
```



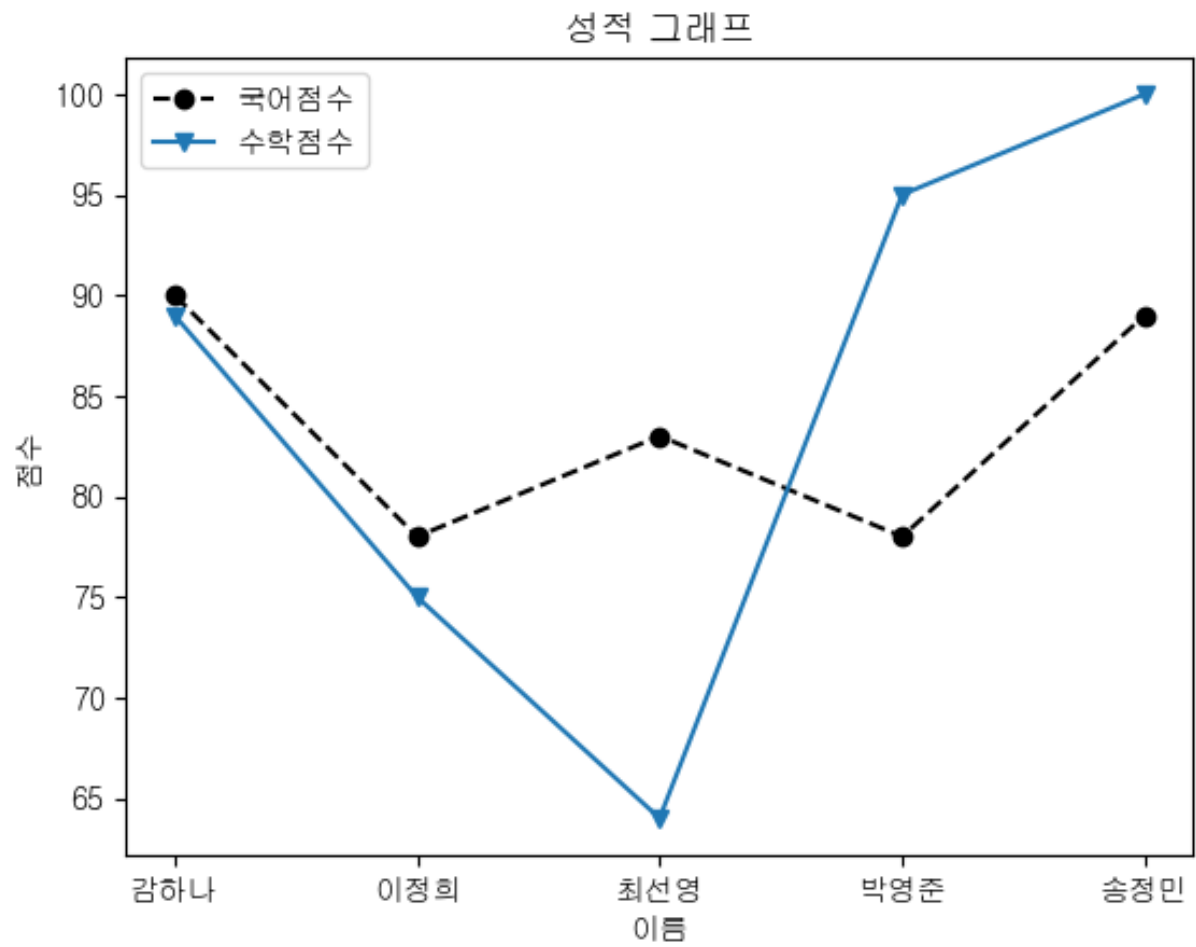
In [2]:

주피터노트북용 :그래프의 선 모양, 색깔, 마커 모양은 아래 코드와 달라도 됨

```
import matplotlib.pyplot as plt
import pandas as pd

plt.rc('font', family='Gulim')
df=pd.read_excel('data1.xlsx')

plt.plot(df['name'], df['kor'], 'k--', marker='o', label='국어점수')
plt.plot(df['name'], df['math'], marker='v',label='수학점수')
plt.title('성적 그래프')
plt.xlabel('이름')
plt.ylabel('점수')
plt.legend()
plt.show()
```

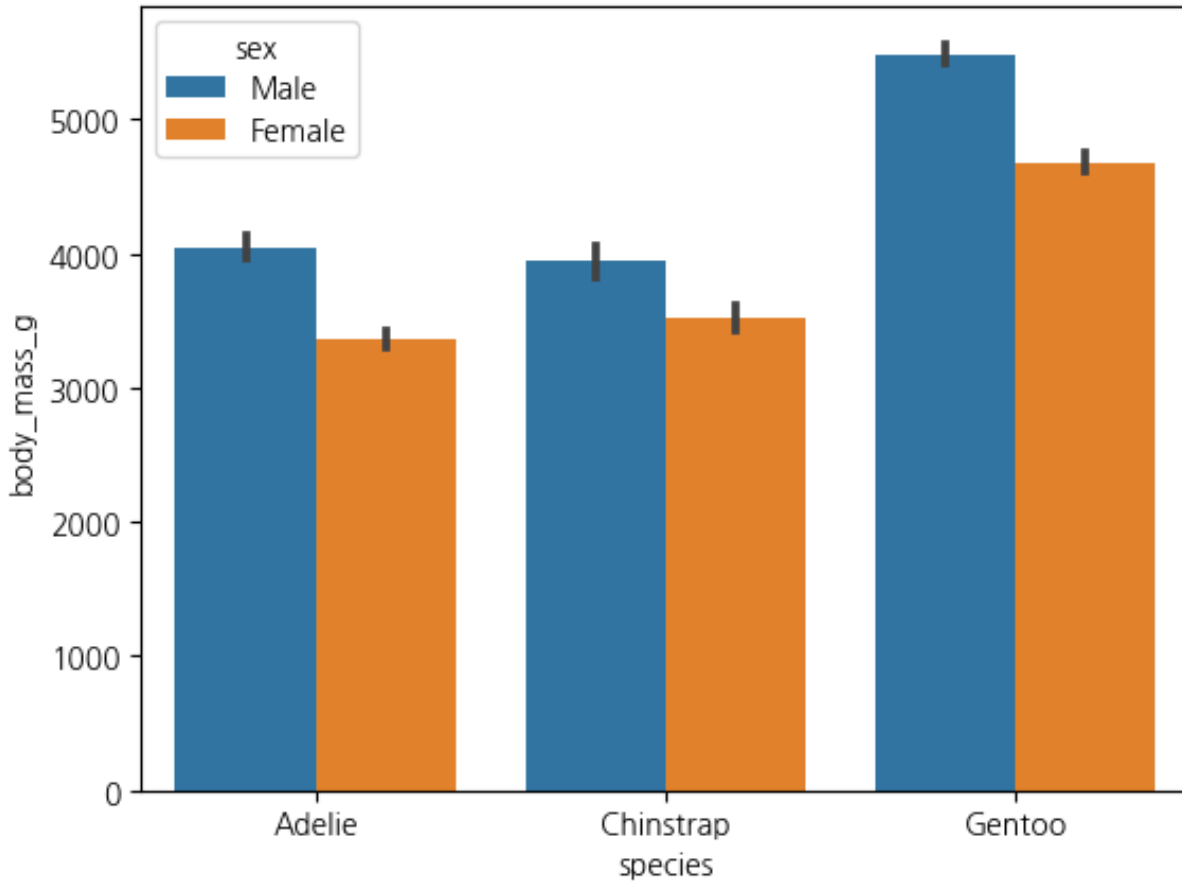


미션3

In [3]:

```
# 미션 3-1)
import matplotlib.pyplot as plt
import seaborn as sns

penguins_data = sns.load_dataset('penguins')
sns.barplot(x = 'species', y = 'body_mass_g', data=penguins_data, hue='sex')
plt.show()
```

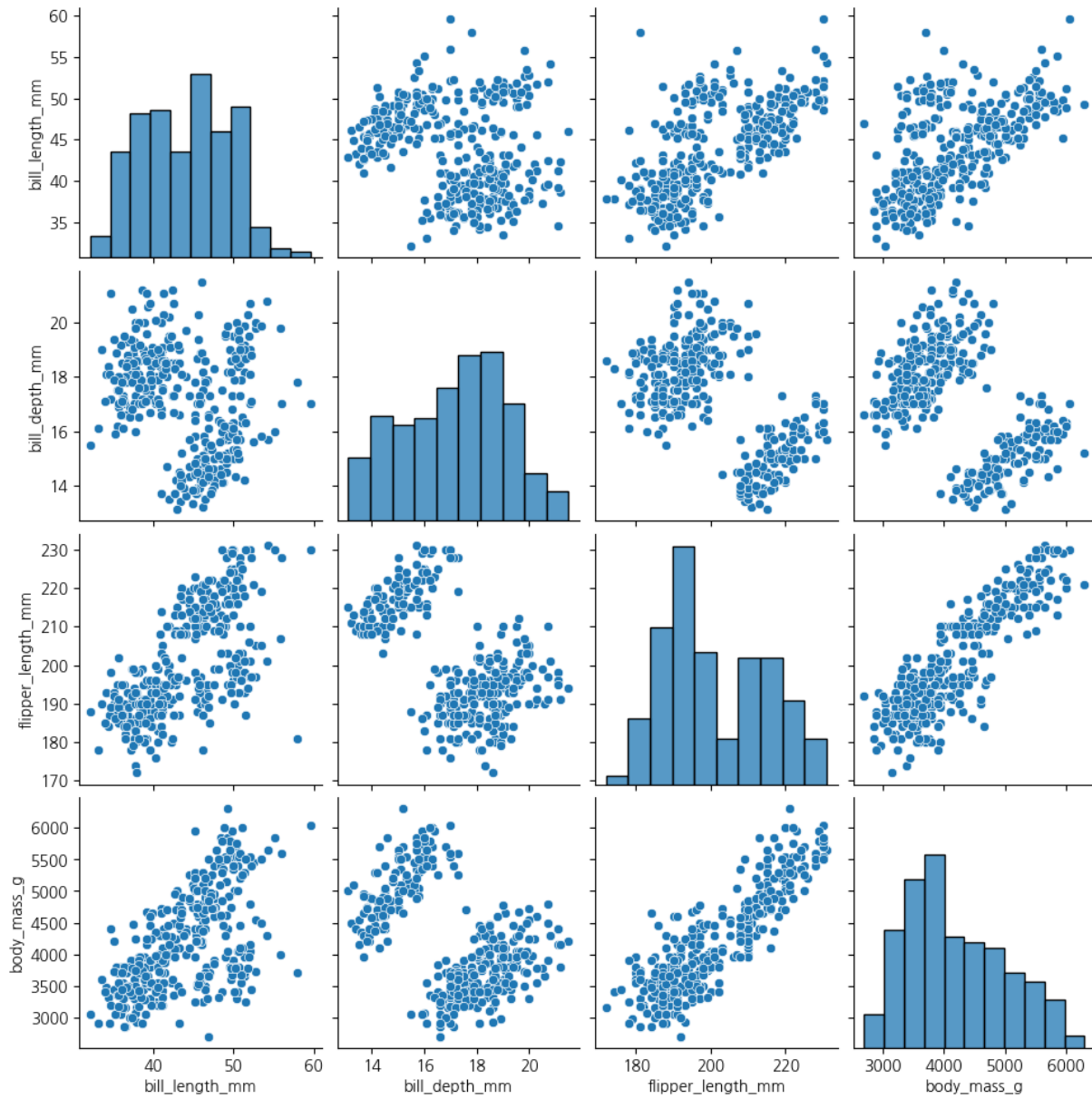


```
In [4]:
```

```
# 미션 3-2)
```

```
sns.pairplot(data=penguins_data)
```

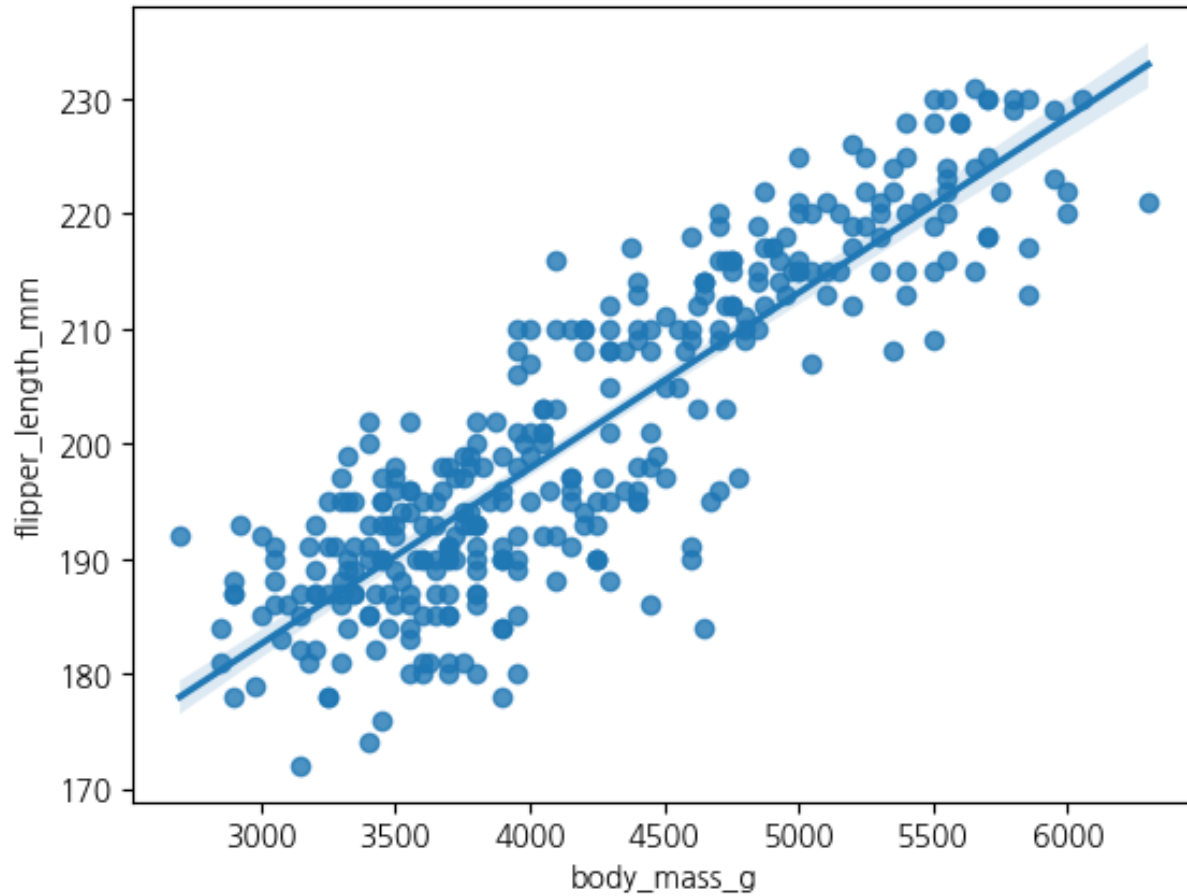
```
plt.show()
```



```
In [5]:
```

```
# 미션 3-3)
```

```
sns.regplot(data=penguins_data, x='body_mass_g', y='flipper_length_mm')  
plt.show()
```



연습문제

In [4]:

```
# 2번 문제
```

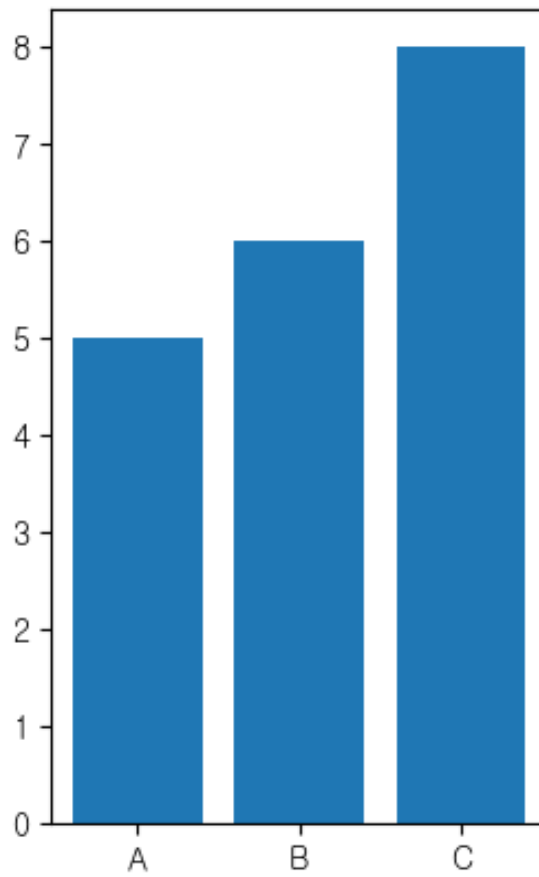
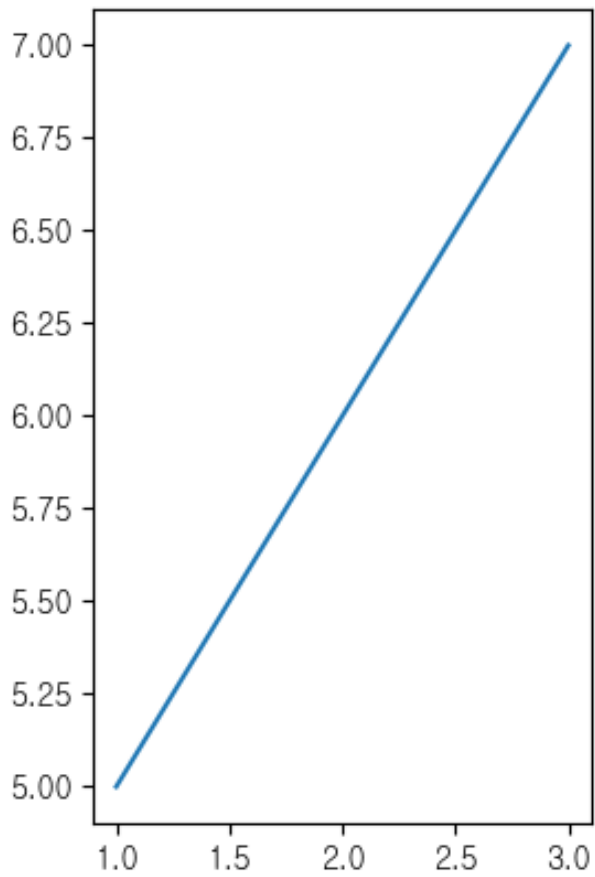
```
import matplotlib.pyplot as plt
```

```
fig, ax = plt.subplots(1,2)
```

```
ax[0].plot([1,2,3],[5,6,7])
```

```
ax[1].bar(['A','B','C'],[5,6,8])
```

```
plt.show()
```



In [5]:

```
#3번 문제를 위한 데이터 가져오기
import matplotlib.pyplot as plt
import seaborn as sns

iris_data = sns.load_dataset('iris')
iris_data
```

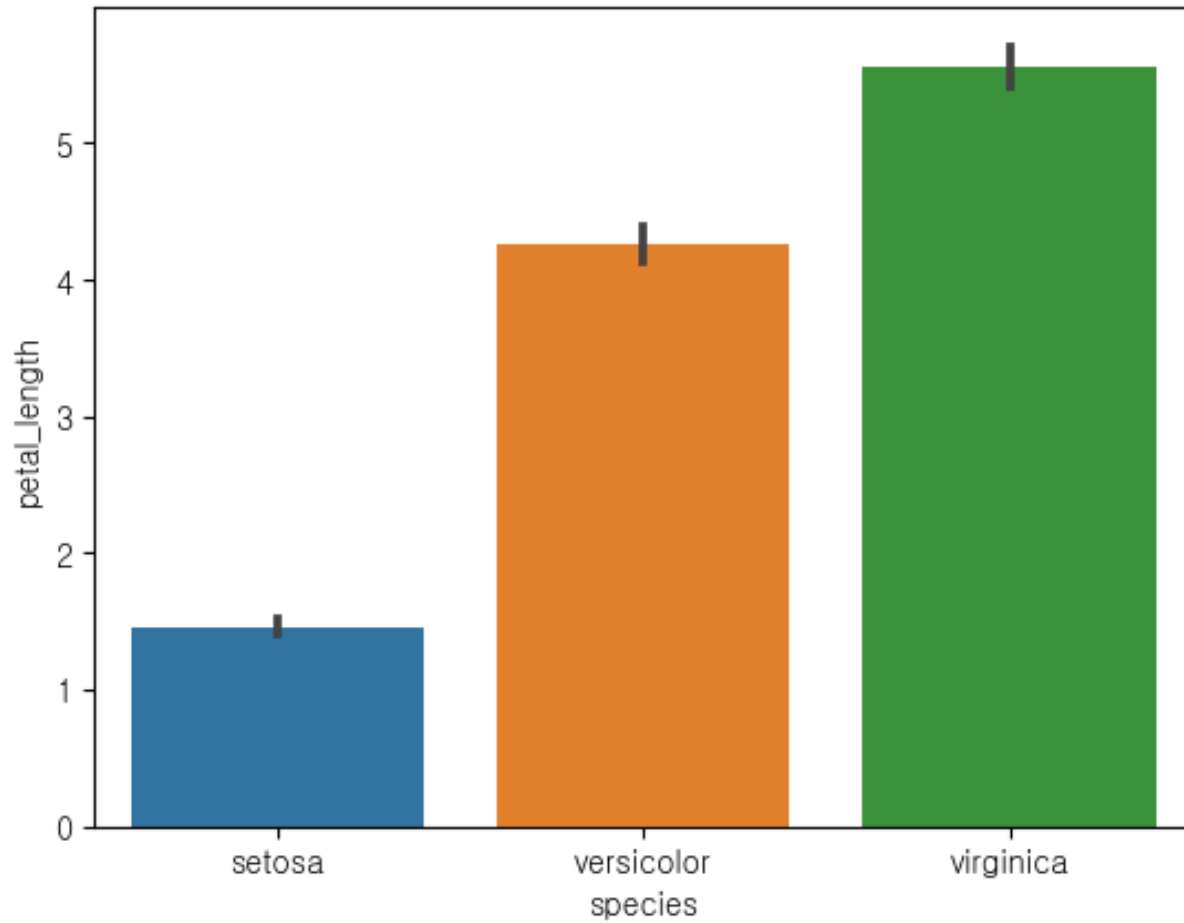
Out[5]:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

In [6]:

```
#3번 문제에서 요구하는 그래프 그리기 : 붓꽃 품종별 꽃잎 길이 평균을 막대그래프로 나타  
sns.barplot(data=iris_data,x='species', y='petal_length')  
plt.show()
```

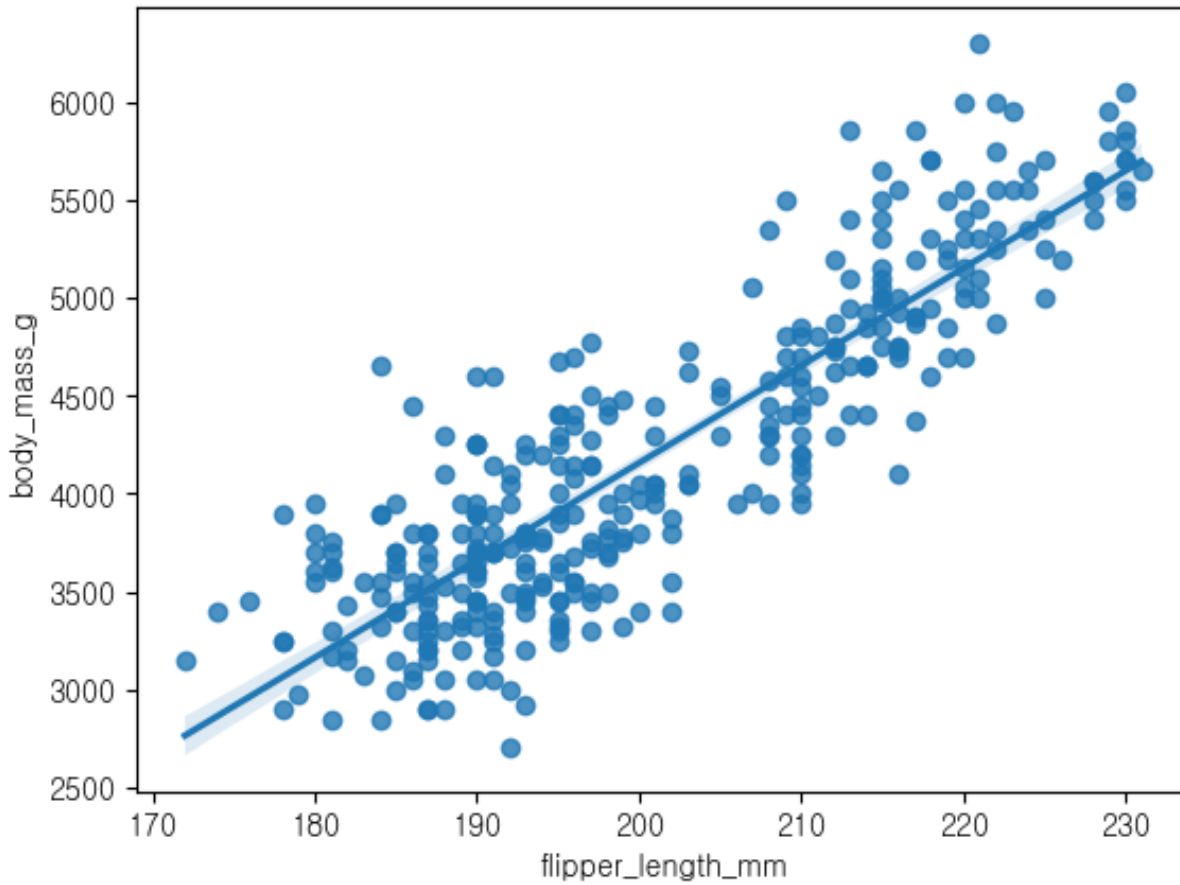


In [7]:

#4번 문제

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
pen_data = sns.load_dataset('penguins')
sns.regplot(data=pen_data, x='flipper_length_mm', y='body_mass_g')
plt.show()
```



In []: